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NFTS AND THEIR POSSIBLE USE IN THE PHYSICAL WORLD

Bachelor Thesis

Study programme: Applied Informatics

Specialization: Applied Informatics

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Prague, February 2023

Acknowledgement

I am grateful to Jan Rejthar for his guidance and open-mindedness as my supervisor. I could not have asked for a better guide. Additionally, I extend my thanks to all the individuals I encountered during my studies.

Abstract

In 2021, the sudden surge in popularity of non-fungible tokens, commonly referred to as NFTs, caused a revolution in the digital art industry and gained a widespread media attention. However, this hype resulted in unrealistic expectations that proved unsustainable, leading to a decline in interest and widespread perception that NFTs were a dead-end. Despite this, new applications continue to emerge, drawing in fresh audiences.

This thesis aims to explore unconventional perspectives and propose potential use cases that demonstrate how non-fungible tokens can enhance our physical world and improve our lives in the future.

Keywords

Non-fungible token, NFT, Blockchain, Cryptocurrency, Use cases, NFT market

Abstrakt

V roce 2021 náhlý nárůst popularity nezaměnitelných tokenů, často označovaných jako NFT, způsobil revoluci v digitálním uměleckém průmyslu a získal rozsáhlou pozornost médií. Nicméně, tato trakce vedla k nerealistickým očekáváním, což vedlo ke snížení zájmu a široce rozšířenému názoru, že NFT technologie je slepá ulička. Přesto se objevují nové aplikace, které přitahují čerstvé publikum.

Tato práce si klade za cíl prozkoumat nekonvenční perspektivy a navrhnout potenciální případy použití, které demonstrují, jak mohou nezaměnitelné tokeny zlepšit náš fyzický svět a zlepšit naše životy v budoucnu.

Klíčová slova

Nezaměnitelný token, NFT, Blockchain, Kryptoměna, Případy použití, NFT trh

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Introduction

The emergence and development of new technologies continue as the world undergoes digitisation. Modern society consistently integrates new technologies like blockchain and cryptocurrency into their daily routines. One of the latest advancements in blockchain technology is the non-fungible or irreplaceable token, commonly known as NFT. These undergoing shifts affected me personally, leading to my discovery of a non-fungible token. Seeing its benefits and potential, I decided to introduce this technology, ending my bachelor's degree with something of my interest.

The objective of this bachelor's thesis is to introduce possible use cases, going beyond the usual idea that a non-fungible token is used only in the digital art industry. The thesis will start by defining the environment where an NFT operates, followed by a comprehensive analysis of the related topics to understand non-fungible tokens. To break unusual perspectives, we will shine a light on many present applications of an NFT, revealing the building process happening behind the scenes. The next important step is to analyse the current market trend, revealing the ongoing situation so that we can forecast and predict future development. Finally, the thesis will introduce possible future applications of non-fungible tokens, with the hope of enhancing our physical world. Each use case will explore current challenges, obstacles, and potential improvements and provide an achievable solution.

Since non-fungible tokens are a young and continuously evolving technology, locating official research papers on the subject can be challenging. As a result, this difficulty is noticeable in thesis citations, where websites are cited more frequently than usual.

1 Centralisation and decentralisation

To understand NFTs (non-fungible tokens) and how they can be used, a few key concepts and underlying technologies will be introduced upon which non-fungible token technology is built. Comprehending the influence and application of the third World Wide Web, the blockchain, and cryptocurrencies will bring us a clear picture of the researched topic.

We are starting with defining three types of systems: centralised, decentralised and distributed. In a centralised system, one authority controls and oversees every activity, institution, or transaction. A government is one of the most known centralised systems (Hackl, Lueth, Di Bartolo, Arkontaky, Siu 2022). On the other hand, a decentralised system takes away control and decision-making from a central entity and puts it in the hands of its users participating in a network (What is Decentralization? [no date]). The main advantage of a decentralised system is that when one server, also called a “node”, goes down or has been attacked by a hacker, it will not affect the whole network like in a centralised one. When this situation occurs in a decentralised network, the affected nodes reconnect to another main network. There is also another type of system called distributed. In a distributed system, there is no single place that processes data. All the files are shared across a network, making the system more resistant to data overload and attaining a quicker response time (Hackl, Lueth, Di Bartolo, Arkontaky, Siu 2022).

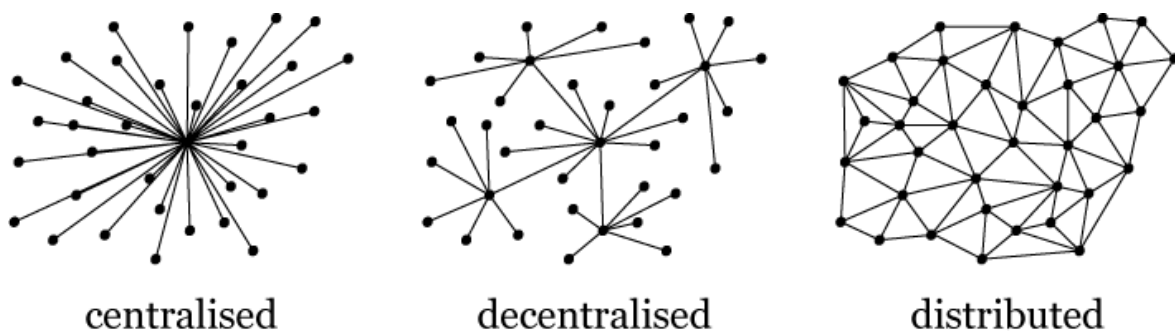


Figure 1: Types of Systems (Phil Windley 2015)

1.1 Web 3.0

In this subchapter, our current transient state of the web will be explained, but before that, we will briefly touch upon the first two stages of web development to indicate the changes made over time. The main point here is to define the environment where a non-fungible token is alive and developing as a new technology.

In the first stage of the web, websites were only readable, and the data was fixed. The second stage is known to be readable and editable. It becomes possible to interact with the websites actively and, more importantly, to create them. This new opportunity was taken by

a few tech giants that now control most of the Internet, providing us with services in exchange for our data. Thus, the current World Wide Web is mainly centralised, where every transaction or action goes through a trusted third party (Ma, Huang 2022).

Web 3.0, a third generation of the World Wide Web, aims to shift ownership of the Internet from Big Tech companies to individual users. The web of the third-generation ultimate vision is that there will be no dominant “mega-corporation”. Instead, it will be built by millions of users, creators, and programmers worldwide (Ma, Huang 2022). With Web 3.0 technology, it will be possible to securely transfer our information, assets, and money without a third party (Ragnedda, Destefanis 2019). The change will be achieved with the principle of decentralisation and blockchain technology’s help.

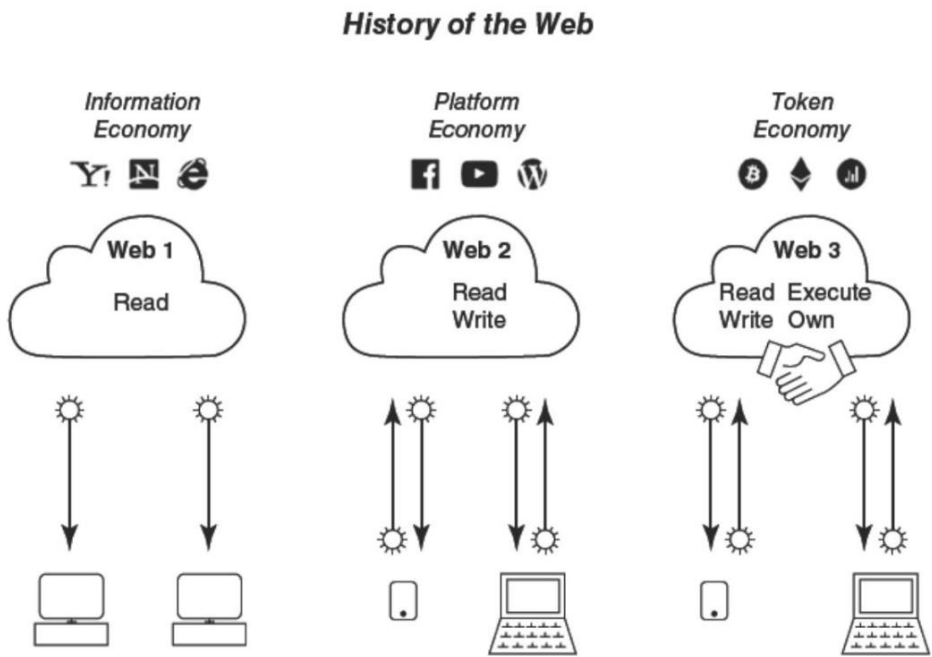


Figure 2: History of the World Wide Web (Ma, Huang 2022)

2 Blockchain

Blockchain is a cornerstone of Web 3.0 and a kernel of a non-fungible token. NFT is built on the blockchain. Without it, a non-fungible token could not exist and function. This chapter will explain what the blockchain is and how it operates—closing with addressing the revolutionary functions that have brought us one particular blockchain.

The critical thing to know about the blockchain is that it is both a decentralised and distributed system (Hackl, Lueth, Di Bartolo, Arkontaky, Siu 2022). Blockchain is a distributed ledger. We can understand this term as a “list of records” or a database shared across a public and private computing network. Each participant in the network or so-called “node” holds a copy of the ledger. Every new information or transaction is mathematically encrypted and publicly added as a new “block” to the chain of records. Before adding a new block, various consensus protocols validate that block with other participants. All nodes then update their version of the blockchain ledger to be identical. This prevents fraud or double spending. In addition, there is no requirement for a central authority because partakers of the network are validators of that block (Ma, Huang 2022). The block contains data, hash and hash of the previous block. Hash can be defined as an imprint of numbers that act as a unique identifier for each block, which makes it possible to access all the previous blocks’ activity from the latest block (Singhal, Dhameja, Panda 2018). That is why it is called the blockchain. “Block” of digital data on a “chain” that connects to other blocks.

2.1 Ethereum

Moving forward, our primary focus will be on the Ethereum blockchain due to its widespread popularity and robust functionality in the realm of NFTs.

A new door was opened by Ethereum, transforming blockchain technology with an abstraction layer. The abstraction layer lets transactions from different participants and applications be generalised to the program code, which is then passed to all the Ethereum nodes. Ethereum enables users to write a script as a smart contract that can virtually do anything from a programming perspective. One can code almost any program that runs on every blockchain network node (Singhal, Dhameja, Panda 2018).

2.2 Smart contracts

Initially invented by the Ethereum blockchain, it is a collection of data and code deployed using cryptographically signed transactions on the blockchain network. All nodes must derive the same results for the execution of a particular contract. Results are then recorded into a block (Ma, Huang 2022). Simply put, smart contracts are digital contracts that automatically execute specific actions once pre-set conditions are achieved.

3 Cryptocurrency

So far, we have some basic knowledge of decentralisation and blockchain technology, thus making us one step closer to comprehending NFTs. This chapter will introduce the implementation of blockchain as a digital currency and how we can later distinguish between fungible and non-fungible assets.

Cryptocurrency is cryptographically secured digital or virtual currency existing only on the Internet. Whatever someone does with a cryptocurrency, the change must be verified on the blockchain as a transaction. Every activity, including purchasing, selling, or the authentication process, leads to the creation of transactions that are grouped into a block by the system. When the block is full, it is confirmed by the nodes, and thus this process makes the blockchain grow longer by linking the newly created block to the previous one. We can think of it as an extremely long accounting book (Terry, Fortnow 2021).

3.1 Coin vs Token

There is an essential distinction between coin and token. Coins as cryptocurrency each have their respective blockchains. A token does not have a blockchain. Thus, tokens utilise another coin's blockchain. If we are talking about Ethereum as a digital currency, it is a coin because it has its own blockchain. Coins are usually used for one thing, payments. On the other hand, tokens are made for numerous purposes. On the Ethereum platform, developers can create smart-contract-enabled tokens which can be utilised with other products and services. Tokens of such represent an asset, cryptocurrency, access, or anything else that is not unique and can be transferred (Terry, Fortnow 2021). In other words, the tokens mentioned here are not non-fungible and have attributes that make them the same as the other tokens, more precisely in value and type.

The process of creating cryptocurrency coins and tokens is called “minting”. The mint term is like minting a physical coin. Developers can choose to set a maximum supply of coins or tokens. In the case of Ethereum, the Ethereum blockchain has an infinite supply. It is built as a platform for creators, utilising the abstraction layer for producing programs that run on a blockchain using Ethereum's virtual currency, unlike Bitcoin, with a predetermined maximum supply of 21 million coins, where the developers' original goal was to make a purely digital currency (Terry, Fortnow 2021).

Fiat currency also has no maximum supply, and a central bank controls its monetary policy. Thus, more and more are printed in the case of dollars, causing the value of each dollar to decrease if the demand for dollars is the same. On the other hand, monetary policy in a decentralised system depends on protocols managed by a decentralised autonomous organisation (DAO), contrasting a central bank with a fiat monetary policy (Terry, Fortnow 2021). A decentralised autonomous organisation is a form of legal structure without a central entity, where decision-making depends on members of the entity. Decision-making

relies on smart contracts, where the outcome of a particular decision can increase the circulating supply or burn some of the tokens (Nathan Reiff 2022).

Burning the token means destroying it. The process sends a token to a null address (digital wallet) to which no one has access, meaning no one has a private key to this address on the blockchain, making it impossible to restore.

3.1.1 Fungible asset

Dictionary.com describes the adjective fungible as “being of such nature to be freely exchangeable or replaceable, in whole or in part, for another of like nature”. More practical would be to say that, for example, dollars are fungible. One person can give someone a five-dollar bill; the other would give him five one-dollar bills in return. It does not matter which bill the other person gave; dollars are interchangeable. The same goes for fungible tokens and digital currency. It does not matter from which wallet anyone sent a Bitcoin. A Bitcoin is a Bitcoin. We can even apply this to some goods such as oil. One barrel of oil of the same grade can be changed between parties, which would not make a difference (Terry, Fortnow 2021).

An example of a non-fungible real-world item is a diamond. Every diamond is uniquely cut and shaped. It is merely impossible to interchange (Terry, Fortnow 2021).

3.2 Crypto wallets

Crypto wallets or digital wallets do not store virtual assets. Instead, they give access to them. Digital assets live on the blockchain, and a digital wallet controls and manages them. On a technical side, crypto wallets keep private keys. Private keys prove ownership of an owned digital asset (Derousseau 2019). Every digital wallet has a unique address, like an email address, thus making it possible to send transactions to different wallets.

There are a few types of crypto wallets. The most secure is a hardware cryptographic wallet that stores assets offline called a “cold wallet”. Then there are many variations of “hot wallets”, meaning your assets are stored online. One can be a mobile or web application enabling interaction with an account directly in the browser (ethereum.org 2023a).

4 NFT (Non-fungible token)

Digital space is growing fast, and there is a demand to replicate somehow the properties of physical items like proof of ownership, uniqueness and scarcity. This is where non-fungible tokens come into place. The current chapter will tell us what makes a non-fungible token non-fungible.

The non-fungible token introduces a crucial element to the Web 3.0 environment: proof of ownership. Individuals and group entities can own parts of the Internet. Thus, NFTs represent ownership of unique items on the blockchain. They are used to tokenise collectables, art, deeds, real estate and more. There is no possibility of altering the record of ownership or making a copy of the existing NFT because any change can be verified and seen on the blockchain (ethereum.org 2023b).

However, what makes each NFT unique? The functions of an NFT are similar to a cryptocurrency. Nevertheless, when minting an NFT via smart contract, each created token has a maximum supply of one, making them wholly uninterchangeable. A non-fungible token cannot be freely exchanged with something the same because there is no identical token (Terry, Fortnow 2021).

We can take a photo, mint it into a token, and put it on the blockchain with a unique identifier. If someone wants to prove ownership of that photo or view transaction history to confirm authenticity, they can easily do so with the owner's wallet address that holds that non-fungible token because the blockchain is an open network (Terry, Fortnow 2021).

4.1 Features of the non-fungible token

There are a few different characteristics that exclusively NFTs have, thus setting them apart from the regular fungible tokens. We will take a look at some of them.

4.1.1 Mandatory aspects

A crucial requirement for every NFT is its *name*. Additionally, the *main content* of an NFT refers to the specific content for which it was created. In the case of digital art NFTs, this would typically be an image, video, 3D model or GIF. In contrast, for NFTs representing digital land, the main content would be the coordinates of the specific location within the virtual world. Lastly, it is important to note that the *supply* of an NFT is always one, as mentioned previously (Terry, Fortnow 2021).

4.1.2 Optional aspects

Unlockable content refers to information exclusively visible to the owner of an NFT, such as login credentials, a game activation key, or a redeem code for a physical item. The

NFT's *description* typically includes a description of the unlockable content, as well as any applicable copyright or trademark notice. Additionally, *ongoing royalties* may be set up for certain types of NFTs, such as those representing digital art. In such cases, the original owner may receive royalties each time the NFT is sold to a new owner. These royalties are automatically sent as digital currency or any other agreed-upon reward (Terry, Fortnow 2021).

While some aspects of an NFT are stored on a blockchain, the majority of its characteristics are included in the metadata of its associated smart contract. Metadata refers to information about other data. For example, the image representing an NFT is not stored in the smart contract itself but is instead saved elsewhere, with a reference to its location included in the metadata of the corresponding smart contract. This approach ensures that the blockchain is not overwhelmed with large files, which can be costly to store. To avoid these costs, most projects store metadata that references other metadata instead of storing it directly on the blockchain (Terry, Fortnow 2021).

4.2 Metaverse

Playing a huge role in Web 3.0, the Metaverse is a conjunction of two ideas that have existed for many years: virtual reality and digital second life. It is hard to define this concept in its early stages. It can be a virtual and augmented reality or a video game. There is even a possibility that it will mirror our physical world (Ma, Huang 2022).

Non-fungible tokens allow for digital ownership within the virtual world, enabling players to possess virtual real estate, clothing, avatars, and more using NFT technology. This is already evident in current virtual world games like *The Sandbox* or *Decentraland*. However, it is important to note that this is still just a concept at present. The true realization of the Metaverse is not yet feasible using existing technologies.

5 Present state of affairs

As of the writing of this thesis in the first quarter of 2023, the most hyped tech buzzword of 2021, non-fungible token technology, has undergone significant changes throughout its existence. Spreading out, finding its application in many different corners of the digital realm, even into our physical world. Unsurprisingly, Web 3.0 solves some of our current problems, and NFT is no exception. This chapter will introduce current NFT use cases to show new ways people apply this technology, bringing in beneficial solutions for people and companies. Most importantly, we will analyse the present market state and determine what to expect from non-fungible token technology, estimating the NFT market trend for the future.

5.1 Current use cases

Art and collectables

For most of us, NFTs are known as digitalised pictures. Any human being can buy one for a couple of dollars, or someone can spend a whole fortune on one digital picture on a blockchain (Ellen Glover 2022). In 2021, NFTs gained widespread popularity as collectables, leading many to assume that their utility was limited to this purpose. However, utilising NFTs as collectables only scratches the surface of their potential use cases.

As expected, the digital art sector has discovered a way to utilise non-fungible tokens. Artists can leverage NFTs to reap numerous benefits. For artists lacking privilege or influential connections, it can be challenging to gain traction. However, by creating an NFT as a means of recording and verifying the authenticity of their work, artists can increase their visibility. Moreover, thanks to smart contracts, artists can earn automatic royalties. The barriers to entry are low, requiring only artwork, a digital wallet, and some tokens. Most intriguingly, artists can cultivate exclusive fan communities, offering members-only merchandise, meetups, or private social media channels for those who have purchased their art as an NFT (Ellen Glover 2022). One of the most popular NFT artists is *Beeple* selling his NFT for \$69.3 million in 2021.

Music

Traditionally when the music artist released a new song or an album, 90% of the revenue went to other parties, such as the producer or a studio. With NFTs, music artists can tokenise their songs and albums, providing royalties to producers and supporters. Blockchain technology is already being used in the music industry by *Grimes*, *the Weeknd* and others. The music industry with non-fungible token technology enables artists to take 90% of the income, and the platform takes 10% (Ellen Glover 2022).

For example, music composer *Nas* used NFT technology to provide royalties based on the streaming income of his song and gave additional physical benefits, such as VIP concert tickets for the holders (Cem Dilmegani 2022, p. 7). Holders are entities that hold an asset for a more extended period of time.

Fashion

The fashion industry is moving towards the Metaverse concept, using current Metaverse platforms with NFTs. In the Metaverse, clothing is a collectable and a way of expression. Avatars will use them to socialise with friends, go to concerts or meet in the virtual world. Major brands such as *Gucci* sold an NFT bag on the *Roblox* gaming platform for over \$4,000. A nine-piece collection of NFTs have been sold for \$5.6 million by *Dolce & Gabbana* (Ellen Glover 2022). A well-known sneaker reseller *StockX* platform uses non-fungible tokens that are tied to a physical product stored in a safe place. A person can buy and resell the digital token or burn the token and receive the physical item (*StockX NFTs - Collect What's Next* [no date]). Moreover, *Prada* enables their holders of NFTs to access exclusive *Prada* events and future drops (Nast 2022).

Gaming

NFTs introduced a new gaming model to the digital space called play-to-earn. Play-to-earn offers real-world economic attraction to the players. In-game items such as avatars, weapons, skins and virtual lands are turning into NFTs which can be taken out of the game to be sold or traded on marketplaces (Ellen Glover 2022).

Many game titles are based on blockchain technology, but the most popular ones are *The Sandbox* and *Axie Infinity*. The *Axie Infinity* game has in-game NFT creatures that can be bred and fight each other. During the playthrough, players can win cryptocurrency tokens and sell them afterwards. During its peak, some players could earn thousands of dollars a month just by playing for a few hours daily. NFTs in gaming can utilise in-game assets for buying, selling and trading (Ellen Glover 2022).

Ticket sales

According to a report released by *Distil Networks* in 2019, bots were responsible for 39 per cent of ticket purchases, which were subsequently resold on secondary markets at inflated prices. This practice not only forces consumers to pay more for tickets but also increases the risk of ticket fraud (Ellen Glover 2022).

As tickets become digital, event holders use non-fungible tokens to create digital tickets on a blockchain. This way, people do not need to use a third party, and a blockchain can easily verify the ticket's authenticity. Creators can set an NFT to be non-transferable, removing resell factor. Moreover, holders of the NFT ticket can get additional benefits such as giveaways, and access to exclusive fan clubs or experiences, all being sent to their crypto wallets (Ellen Glover 2022).

Real estate

Here, we are currently distinguishing two areas where the non-fungible token technology is used. One of them is virtual real estate. These virtual lands exist in the Metaverse already. After purchasing a digital land, the ownership is transferred to the buyer's crypto wallet as an NFT. The largest purchase was made in the Metaverse, *the Sandbox*, for \$4.3 million. Big brands such as *Samsung*, *Gucci* and *JP Morgan* have already bought virtual properties (Cem Dilmegani 2022). The landowner can open a virtual concert venue, build a house, and rent it. The possibilities for monetisation are limitless (Ellen Glover 2022).

The second area, our physical world, is already influenced by NFTs in the form of fractionalised real estate by a company called *RealT*.

Sport

The sports industry is actively using blockchain technology for a new way of monetisation. These one-of-a-kind tokens represent trading cards, sports events or digital memorabilia. Many fans are willing to spend thousands of dollars for a limited edition virtual memorabilia of their favourite player. For example, the lowest asking price for Ja Morant's dunk series 1 is \$475,000 on the NBA Top Shot marketplace (*Impact Of NFTs In Sports Industry And How Are NFTs Being Used In Sports?* 2023).

Physical items

The proof of authenticity feature can be leveraged for luxury items, utilising NFT technology to verify their legitimacy. Various websites currently allow consumers to exchange an NFT for a physical product, providing an additional layer of authentication. Additionally, individuals can create NFTs for their college degrees or certificates on *degreecert.com* (ethereum.org 2023b).

NFTs can also function as an access pass to exclusive restaurants in the real world. There is already one called *Flyfish Club*. NFT holders have unlimited access to a private location with bars, restaurants, outdoor space and social events (Ellen Glover 2022).

As evident, there is a growing trend of discovering innovative applications for non-fungible tokens, expanding their use cases considerably. This highlights how technology continues to progress and evolve as we near the horizon of a new technological era.

5.2 Newly emerging technology

A new emerging technology is always interesting to observe, especially something this practical. In this subchapter, we will analyse the current state of the NFT market and forecast the future model. The results will be applied to existing adoption models to see at which stage non-fungible token technology is and where it is heading.

Each newly emerged technology or even a startup has its patterned lifecycle it goes through to become mainstream or fail on its way. It goes through maturity rotations, and we can somewhat predict the future value of that newly unfolded technology. A few models depict the evolution cycles of newly emerged tech. One such is the Gartner Hype-Cycle model developed by the *Gartner* group, which provides a graphic representation of the maturity and adoption of technologies. It was created to understand how technologies come onto the scene, grow and reach a maturity status, revealing to us on its way risks and opportunities.

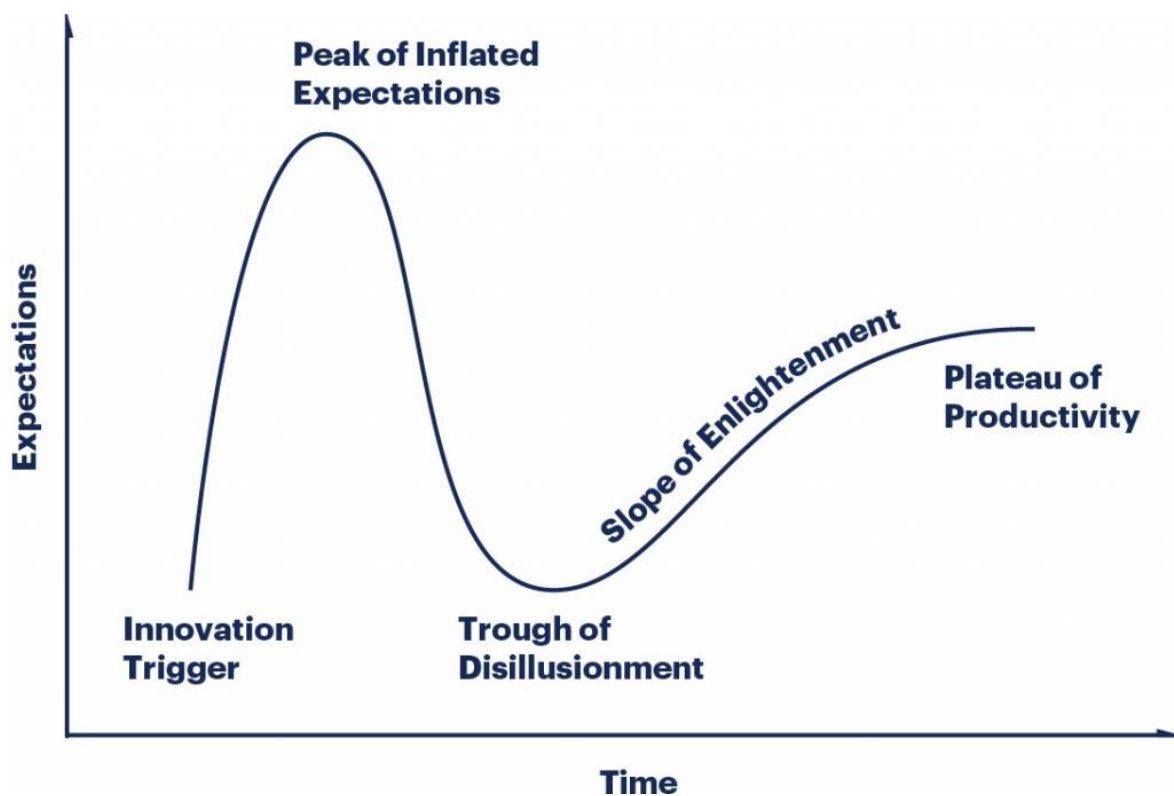


Figure 3: Gartner Hype-Cycle Model (Raza 2020)

The model is divided into five phases. The initial phase, called Innovation Trigger, begins when people hear about new technology on the media or the Internet. Early adopters investigate it, and factors such as high prices and expectations rise. The next one, the Peak of Inflated Expectations, causes technology to be widely publicised. This phase often involves a surge of popularity, flooding the market with products. The next step is the Trough of Disillusionment. Organisations investigate meaningful value beyond initial success stories because technology fails to meet high expectations. Adoption is slowed down, and the market questions the potential of the tech. It is known that many technologies end in the Trough of Disillusionment phase, disappearing or being replaced by another. The Slope of Enlightenment occurs when technology undergoes improvements in both success

and failure stories. At the end of this phase, most late adopters and doubters are yet to adopt the technology shortly. Technology has become mainstream to introduce off-the-shelf solutions, including services and products. This last phase is called the Plateau of Productivity (Raza 2020).

The second model is called Technology Adoption Life Cycle. It is a sociological model which takes a viewpoint from the psychological aspects of defined adopter groups. As shown in Figure 4 (Caio Felipe Santos Matos 2021), the adoption is divided into the early and mainstream markets. In the early market, there are Innovators. These people are technology enthusiasts, the first group to invest in new technology, serving as a test group. Early Adopters are visionaries who serve as an indicator for the rest of the population. To prosper, technology must win over the mainstream market, starting with Early Majority. They want strong practicality and well-established references. Late Majority are waiting until the technology becomes an established standard, investing at the end of a technology life cycle. Finally, there are Laggards, who are sceptical and avoid new technology altogether. Between the early and mainstream markets is “The Chasm”. Crossing the Chasm is the most crucial step to becoming a standard (Sridharan 2017).

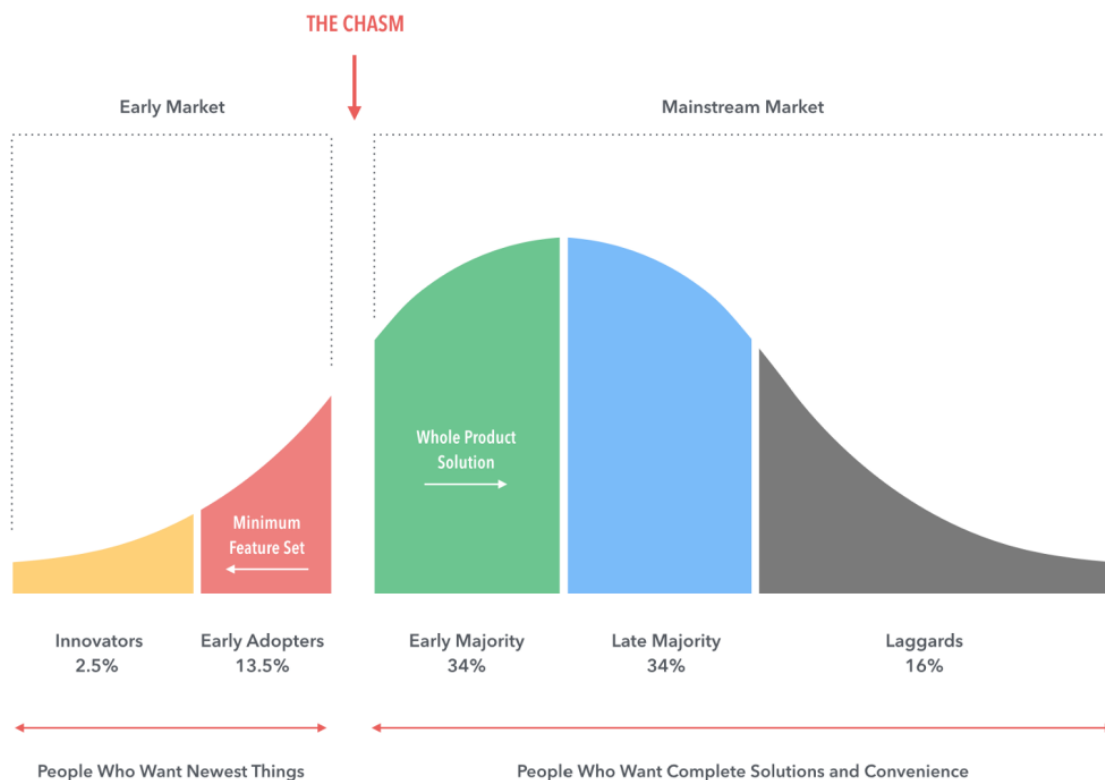


Figure 4: Technology Adaption Life Cycle Model (Caio Felipe Santos Matos 2021)

A non-fungible token is still in its maturity cycle, but where exactly that is the question. With software called *RStudio* which enables us to work with the programming language *R*, we can calculate the current model of the NFT market, and we can go even further and predict the future development of NFTs to show approximately where a non-fungible token technology is headed. For the NFT market, one of the elements that can depict the present hype state is the daily number of transactions, which represents the number of operations daily—portraying people and companies’ interest in this new

technology. The dataset that will be used contains daily transactions on the Ethereum blockchain. After importing our time series and running the script in *R*, which will be attached to the thesis, we get this data visualisation, as seen in Figure 5.

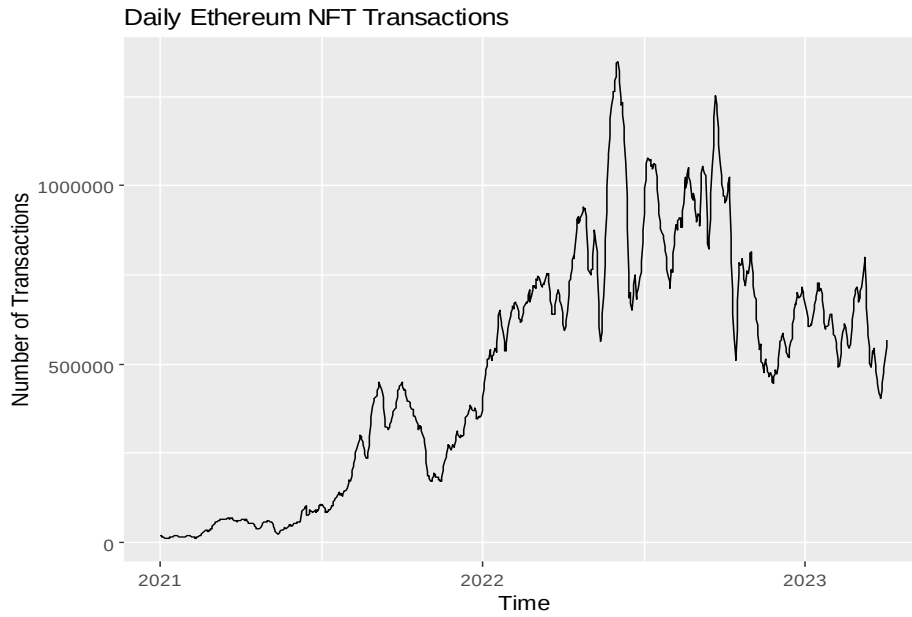


Figure 5: The Current NFT Market Model (Source: author)

If we compare the Gartner Hype-Cycle model in Figure 3 with our current non-fungible token model in Figure 5, we can see that the NFT market has already overcome the Peak of Inflated Expectations phase and is now heading down. During the Peak of Inflated Expectations phase, the number of transactions exceeded the one million mark, reaching its all-time high. The fall of cryptocurrency prices is related to the NFT market crash but is not the only factor. The hype around profile pictures NFT collections could not last forever, and the combination with the cryptocurrency market plummet led to the market collapse. We cannot say it is good or bad, but it can and is transforming how non-fungible tokens are used, which we can see already happening with newly emerged use cases. People lost interest, but companies started to build and develop off-scene. From the current model, we are still determining whether the NFT market has already hit the bottom of the Trough of Disillusionment phase and is ready to steadily grow to the Slope of Enlightenment phase or is only on its way there.

To eliminate uncertainty, we will use our script to forecast the possible direction of the non-fungible token market. The *Prophet* software will build the forecasting model, which decomposes the time series into pieces, predicting “Signal” and adding “Noise” or simply confidence intervals for uncertainty like any other prediction model. However, every model has a different approach to predicting the Signal. In the case of *Prophet*, this software breaks Signal into Growth/Trend, Season and Holiday. It is a curve-fitting approach to

forecasting. Using the *Prophet* software in *RStudio* at our current time series, we get the predicted model depicted in Figure 6.

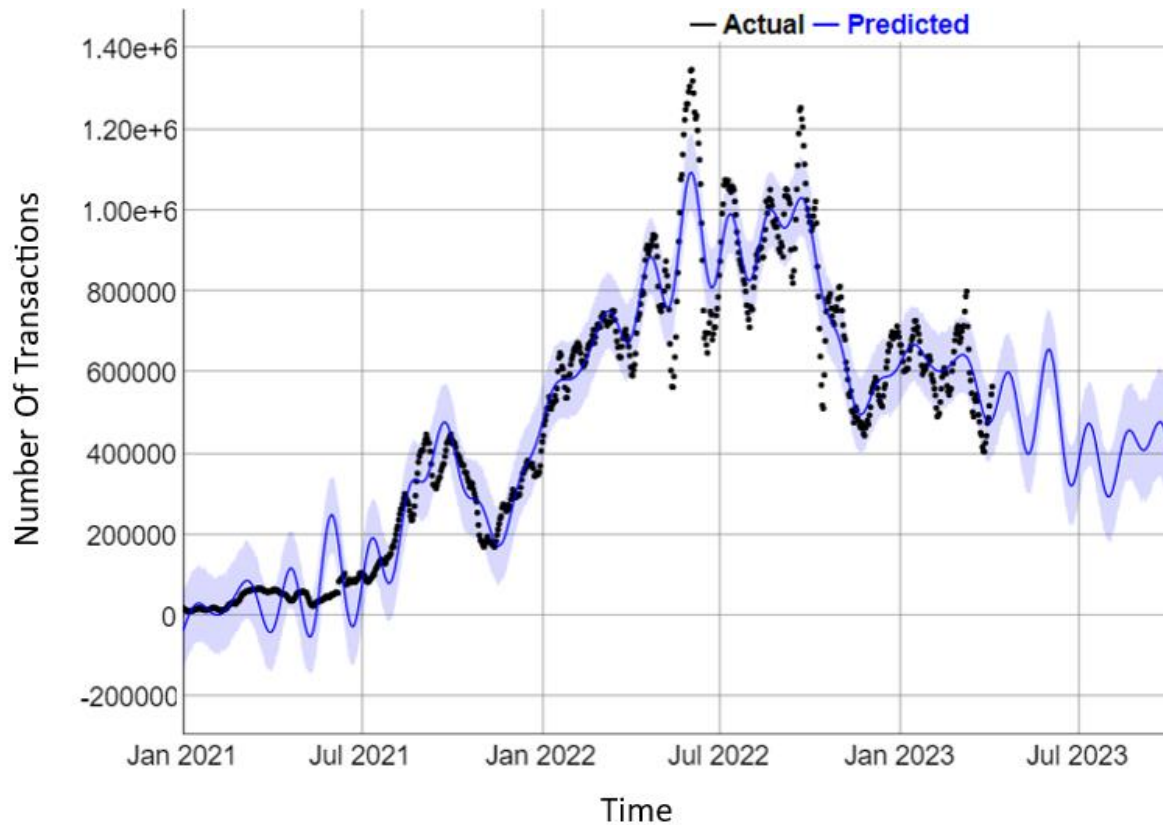


Figure 6: Predicted Model (Source: author)

Let us break down the model to understand it better. On the horizontal axis is time, and on the vertical one is the number of transactions. Black dots are actual existing data from the time series we imported. The predicted data are set for the next half a year, represented by a blue line. This is the Signal. Around the blue line is a light blue area. This area shows the forecast model's error rate, known as Noise. The blue line shows that it is most likely that the NFT market did not reach the bottom of Trough of Disillusionment. The trend will probably decrease till mid-2023, steadily producing around four hundred thousand transactions daily. From there, if the technology wants to pass to the next phase, the number of daily transactions needs to increase to between six hundred thousand and eight hundred thousand. We can even see that after reaching the lowest predicted point on the blue line, there is an increase in transactions, which may indicate the final moments of Trough of Disillusionment and the beginning of the Slope of Enlightenment phase.

To investigate the current state of the NFT market more deeply, we can merge both Gartner Hype-Cycle and Technology Adoption Life Cycle models. This will show us further insights into our researched topic.

If we now set side by side the current NFT market model visualised by the *R* script in Figure 5 and two merged adoption models seen in Figure 7, we can probably say that the market has conquered “The Chasm”. This is evident by the fact that many companies are starting to use non-fungible tokens beyond the art industry, introducing to people new ways of application. Looking at the predicted model, we can say: “There is a large probability that

the market already experienced the needed fall to cross The Chasm and is now finishing the Trough of Disillusionment phase". The predicted model states that popularity is decreasing, but companies are working with the technology. This indicates that the current NFT market is getting traction with an increasing number of Early Adopters, which can lead to entering the mainstream market.

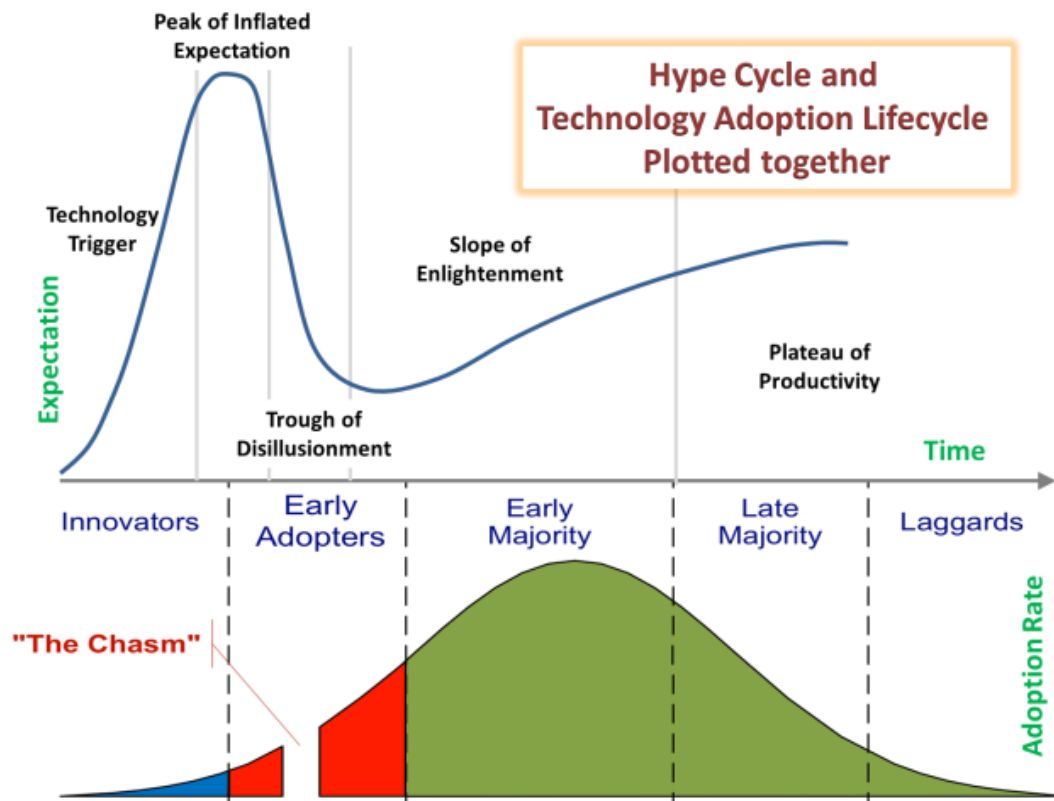


Figure 7: Hype-Cycle and Technology Adoption Models (Udayan Banerjee 2012)

Gartner group stated that it takes approximately 3-5 years for a new technology to go through five phases, but each technology progresses differently. Some can stay at one phase longer than others or go through the maturity lifecycle faster. If the promised value was unrealistic, the technology could disappear, failing to live up to its hype. Some can go back and forth between the phases, as it keeps presenting new values on their way (Raza 2020). In the case of NFTs, they bring something new to the table, dealing with many current problems as our world gets digitalised further still. Thus, this technology will likely stay with us and improve our lives, but first, it needs to overcome some obstacles.

We can assume that it is a perfect time for non-fungible token technology to be considered in some unexplored sectors. Many possibilities and opportunities can arise, leading to significant changes in our digital and physical worlds.

6 Possible use cases

There will be many new use cases in the future, but the range of the bachelor's theses is short, so only one particular sector will be explored that could use NFTs in the long run. The last chapter will introduce three possible use cases.

The government sector is exerting considerable effort to impede the mainstream adoption of blockchain technology, as evidenced in specific countries. Governments in nations like China, Egypt, Iraq, Qatar, and others have banned cryptocurrencies, resulting in a significant setback for the blockchain industry. Several governments are intentionally stalling the implementation process to prolong their control, but only some countries resist change, as we will see later.

However, with technological advancement, there will come a time when most of the government sectors will fully embrace blockchain and NFT adoption. The integration of Web 3.0 is already underway, and to keep pace with the private sector, governments require decentralised systems and their advantages. The critical question is: Can non-fungible tokens be used to benefit the government and its citizens? In this regard, let us delve deeper into a specific use case for the public sector, namely, whether NFTs can be utilised by governments to establish digital identities for citizens operating on a blockchain.

6.1 Digital Identity for Citizens

In each subsequent use case, we will examine the current issues that impact us and the obstacles that affect the NFT market. We will also present the benefits governments can derive from migrating to blockchain technology and non-fungible tokens. Each use case will conclude with a potential solution that involves the use of NFTs.

Identity makes somebody a distinct human entity. Thus, it refers to an individual but can be applied to an organisation or authority. Identifiers are pointers to a particular identity: name, mobile number, date and place of birth, social security number or digital identification credentials. All of those are held and controlled by central entities (ethereum.org 2023c).

Attestations differ from identifiers in that they involve one entity making a claim about another entity. An attestation concerns a claim of an attribute linked to a specific identity. A driver's license is a prime illustration of this, as it contains both identifiers and authorises the holder to operate a vehicle legally (ethereum.org 2023c).

We naturally determine that a digital identity arises from using the personal information on the web and the individual's actions online. A digital identity can be a pseudonymous profile connected to the device or a randomly generated unique ID. Usernames, passwords, health records, driver's license numbers, and online purchasing

history all form a digital identity (*Blockchain for Digital Identity | Real World Blockchain Use Cases* [no date]).

If we talk about decentralised identifiers, the main difference is that they are not managed, issued, or controlled by any central entity. We can create as many Ethereum accounts as possible, and we do not need permission from anyone, and there is no need to store them on the central registry. The result is unique, available and cryptographically verifiable. Attestations come in place as cryptographically verifiable claims made by the issuer, and the Ethereum blockchain acts as a global directory. Decentralised identifiers are secured with private and public keys stored on a digital wallet. This kind of infrastructure eliminates the requirement for certificate authorities (CA). Instead, users will verify digital certificates or encrypt data (ethereum.org 2023c).

6.1.1 Present problems

Most today's countries still use administrative systems from the 19th century, being centralised, hierarchical, bureaucratic systems. In modern times this type of system is outdated and cannot keep up with our rapidly changing world. This leads to growing dissatisfaction and slow administration. Digital Government promised to solve most of the issues. However, centralised databases and insecure data sharing did not provide a level of protection and satisfaction. The blockchain-based government could provide secure storage of citizen and business data. In our times, we cannot even be sure who can access our personal information. Here is a shortened list of significant hacks of the government system in the U.S.:

- In 2011, the Texas Comptroller's Office announced a breach of 3.5 million personal information, dates of birth, and driver's licenses.
- In 2015, the OPM (U.S. Office of Personnel Management) indicated intrusions that impacted 21.5 million people.

As previously mentioned, blockchain provides no single point of failure, and hackers would have to hack every node in the network, which is impossible. The blockchain can do more than protect the data provided to the government. It can reduce high-cost labour processes by using smart contracts with process automation and secure data wallets. Thus, self-generating accountable records, in the form of non-fungible tokens, without a need for multiple levels of oversight and control. Furthermore, it can help combat the lack of trust in government that has been steadily decreasing for many years (Dr. Victoria Adams 2019).

6.1.2 Obstacles

As was stated earlier, the NFT market has to overcome some obstacles to become mainstream. Here we are talking about what needs to be changed for the NFT market to become more relevant and needed in our daily lives, thus, gaining hype and reaching the next maturity cycle. The first is interoperability. Not every non-fungible token is on the Ethereum blockchain, which does not allow them to share security and scalability. This problem is being addressed by the Polkadot network, which is like the "road between all these cities". Storage is a second obstacle. The thesis stated earlier that data are stored off-

chain as metadata. These links point to centralised storage like *Google Cloud* or *Amazon Web Services*, which pose many risks. There is already a peer-to-peer storage system called IPFS, but it needs to gain traction. Moreover, to show that NFTs are more than profile pictures, the accessibility needs to improve so more investors and people can gain awareness of what NFTs genuinely are (Isabelle Lee 2021).

6.1.3 Improvements

Identity inaccessibility

Blockchain and NFTs can solve one of the biggest problems in our society, which is that approximately 1.1 billion people have no proof of identity. The roadblocks are paperwork processes, lack of access, expenses and basic knowledge about personal identity. Someone who does not own a physical identity cannot go to school, apply for a job or get a passport. Interestingly, 60% of 2.7 billion unbanked people own mobile phones. This signifies a robust mobile blockchain-based identity solution for citizens (*Blockchain for Digital Identity | Real World Blockchain Use Cases* [no date]).

Fraudulent identities

There is no standardised way for identification on the Internet. Users use various identities across multiple websites, and the weak link between physical and digital identities creates fake identities (*Blockchain for Digital Identity | Real World Blockchain Use Cases* [no date]). NFTs could solve the issue with bots, where only people with blockchain-based identities could participate in these sales, eliminating the matter in question.

Record management

Government is responsible for keeping up people's records, such as property exchange, passing dates and birth. All of these data could be secured by blockchain, as stated earlier. Many city and state governments got interested in this kind of solution, such as Dubai or the states of Vermont and Delaware which are planning to use blockchain and NFTs for property ownership records, permits, and utility bills (Attaran, Gunasekaran 2019).

6.1.4 Achievable solution

The solution will be centred around accounts (wallets) holding non-transferable "soulbound" tokens (SBTs). They are essentially NFTs which cannot be transferred. SBTs would represent credentials, affiliations and commitments, which function like an extended resume, attested by other wallets with social relations to the object (Weyl, Ohlhaver, Buterin 2022).

The first step is to create wallets called "Souls". They would hold publicly visible, non-transferable (but possibly revocable-by-the-issuer) tokens or "Soulbound Tokens" (SBTs). We can imagine SBTs representing credentials, and memberships issued by other

Souls, being individuals, companies or institutions. A university can be a Soul that attests SBTs to graduates. A family name would be a membership SBT to a particular family, and parents could gift a newborn with SBT. Loans and credit lines could be non-transferable but revocable SBTs. When someone repays, the token is burned and replaced with proof of repayment (Weyl, Ohlhaver, Buterin 2022).

How to not lose your Soul? One solution is to tie Soul Recovery to a Soul's memberships across communities. Communities like employers, colleges, and clubs are considered off-chain; others, like participation in a protocol or a DAO, might be more on-chain. Consent from a qualified majority of a Soul's communities would be needed to recover your Soul's private key. Employers and colleges can burn and re-issue tokens because they have a relationship with someone, making it easy for them to check phone calls or meet with that person (Weyl, Ohlhaver, Buterin 2022). The community recovery is represented in Figure 8 (Weyl, Ohlhaver, Buterin 2022).

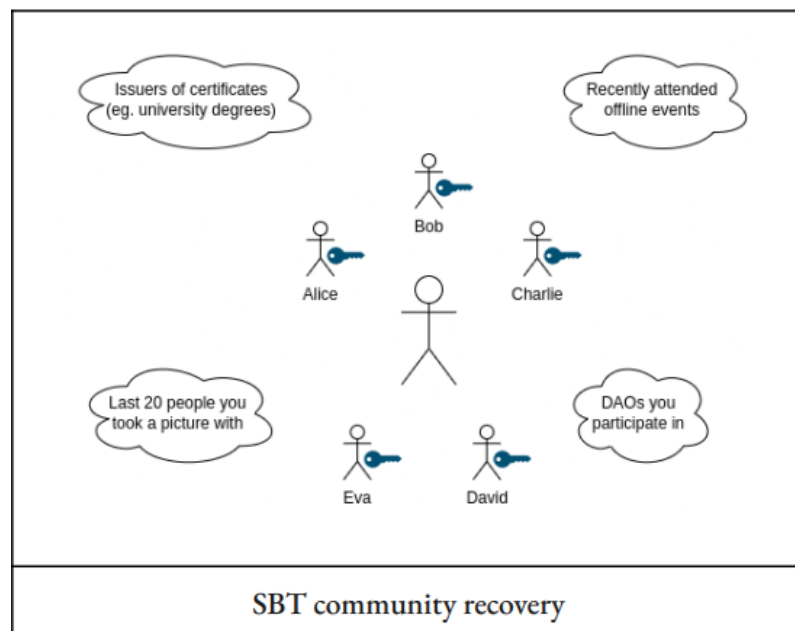


Figure 8: Recovery Process (Weyl, Ohlhaver, Buterin 2022)

“Self-sovereign identity” views data as private property. Only the owner can reveal or keep their personal information secret. However, we can view it as a programmable, loosely entwined bundle of rights to permit access or alter information. Issuers could choose whether to make their SBTs wholly public. SBTs, such as a passport or health record, could be made public from the side of a Soul. This new future digital society would be called a “Decentralised Society” (DeSoc). In DeSoc, Souls are agents, whereas Web 2.0 views Souls (people) as objects. (Weyl, Ohlhaver, Buterin 2022).

One minor problem emerges: blockchain systems are public, and any record is visible to anyone worldwide. The simplest solution would be to store data off-chain leaving only the hash of the data on-chain, illustrated in Figure 9 (Weyl, Ohlhaver, Buterin 2022). It can do so if a Soul wants to make their information public (Weyl, Ohlhaver, Buterin 2022). Another way to define a hash is that it is a converted value derived from a string of characters.

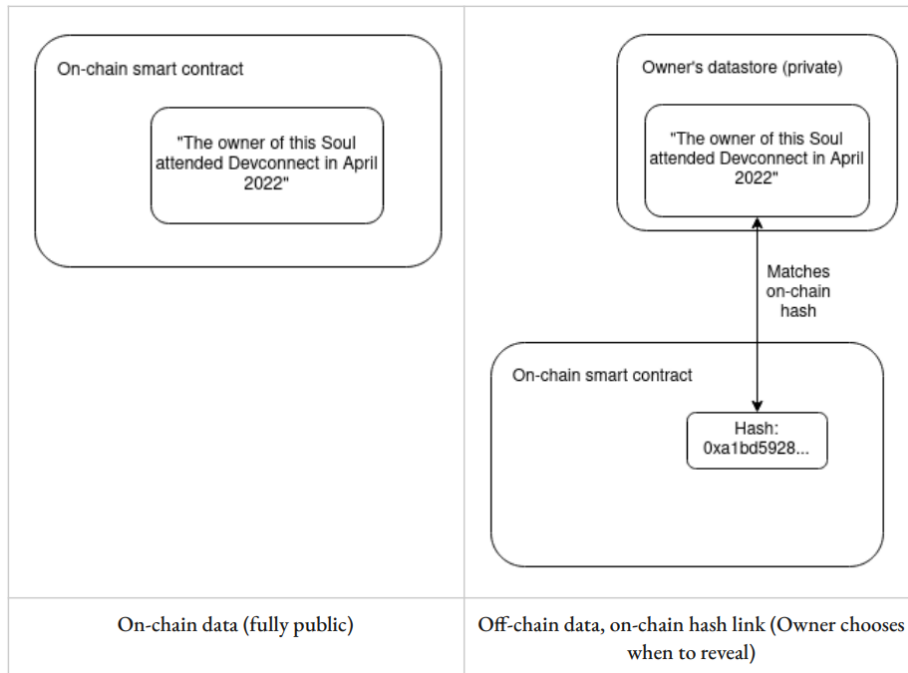


Figure 9: Privacy Solution (Weyl, Ohlhaver, Buterin 2022)

6.2 Voting

In order to establish a connection with the preceding use case, let us consider the potential application of NFTs in voting processes. It is worth noting that the voting use case is closely linked to digital identity, which we explored earlier.

To achieve proficient and successful policymaking in the realm of governance, it is essential to ensure transparent citizen participation in the decision-making process. One of the primary obstacles confronting governance is the issue of voting fraud, which pertains to any illegal actions that occur during the voting period and lead to misleading results. Furthermore, the lack of a transparent authentication mechanism for voting means that citizen participation in governmental proposals lacks transparency (Musamih, Dirir, Yaqoob, Salah, Jayaraman, Puthal 2022).

A verified smart contract can issue NFTs to each participant in an electronic voting system, thus providing them with a distinctive identity. Only those who hold these NFTs will be eligible to participate in the voting process (Musamih, Dirir, Yaqoob, Salah, Jayaraman, Puthal 2022).

6.2.1 Present problems

Consider the national elections in any given country. These events involve thousands of electronic devices running on various software types and networks. Due to their diverse nature, these devices are highly open to hacking and once breached, the hacker can remotely control the device and perform any action. It is important to recognise that all conventional

voting systems are centralised, meaning that if a single function fails on one device, it can cause issues for every other device and system on a network (Gondek [no date]).

In the present era, governance involves numerous departments, resulting in a complex environment where multiple department heads must collaborate to create a regulatory framework. Voting systems, particularly the terminals, are expected to comply with an extensive set of regulatory statements, leading to a clumsy process. This becomes even more challenging when dealing with national elections, as each state requires customized regulatory clauses. This not only creates a tedious management process but also affects provenance (Gondek [no date]).

Therefore, it is more practical to leverage the strengths of blockchain technology. Any new regulatory requirements from any department or state can be easily integrated into the overall system without recoding the logic and halting the entire system. Additionally, this approach would allow stakeholders to track the provenance of each decision, including who overrode a clause, the reason, the timestamp, and the outcome (Gondek [no date]).

6.2.2 Obstacles

The false image of blockchain and NFTs is still present at the moment. People judge technology based on cryptocurrencies and financial sectors. In addition, keep in mind that decentralised systems value all users' anonymity, which can make the attacker undetectable, but this scenario needs to be analysed deeper.

6.2.3 Improvements

Decentralised database

Decentralised blockchain-based systems store voters' credentials securely, mitigating the risk of a single point of failure. Even if a hacker compromises one node, they must breach all others to take down the entire network (Gondek [no date]).

Security

Each voter is provided with two unique keys: a public key that serves as their identity on the network and is visible to others, and a private key that remains confidential and is solely used by the voter to authenticate their identity and execute various operations on a network. The private key can be considered the user's cryptographic signature, and any outgoing transactions cannot be attributed to them without it (Gondek [no date]).

6.2.4 Achievable solution

Figure 10 (Musamih, Dirir, Yaqoob, Salah, Jayaraman, Puthal 2022) depicts a potential scenario in which NFTs can be utilised to ease a governmental voting process. To begin, the government creates and deploys a smart contract with a unique address, and citizens can own a distinct NFT that sets them apart from one another. In addition, the

smart contract can be configured to make the NFTs non-transferable, guaranteeing they stay in possession of their initial owners (Musamih, Dirir, Yaqoob, Salah, Jayaraman, Puthal 2022).

Next, whenever the government proposes a new procedure that necessitates the NFT holders' votes, the details are uploaded to off-chain storage, such as IPFS, to prevent large-sized files from being stored on the blockchain. The voting process can be carried out either on-chain or off-chain. If on-chain, a smart voting contract can be utilised to record all participants' votes, which can be audited transparently and reliably. Voters' signatures can be obtained off-chain by having them sign their vote with their private keys, enabling the authors to verify the legitimacy of a vote (Musamih, Dirir, Yaqoob, Salah, Jayaraman, Puthal 2022).

Finally, voters can delegate their votes to someone else if they are not confident enough to understand the made proposal (Musamih, Dirir, Yaqoob, Salah, Jayaraman, Puthal 2022).

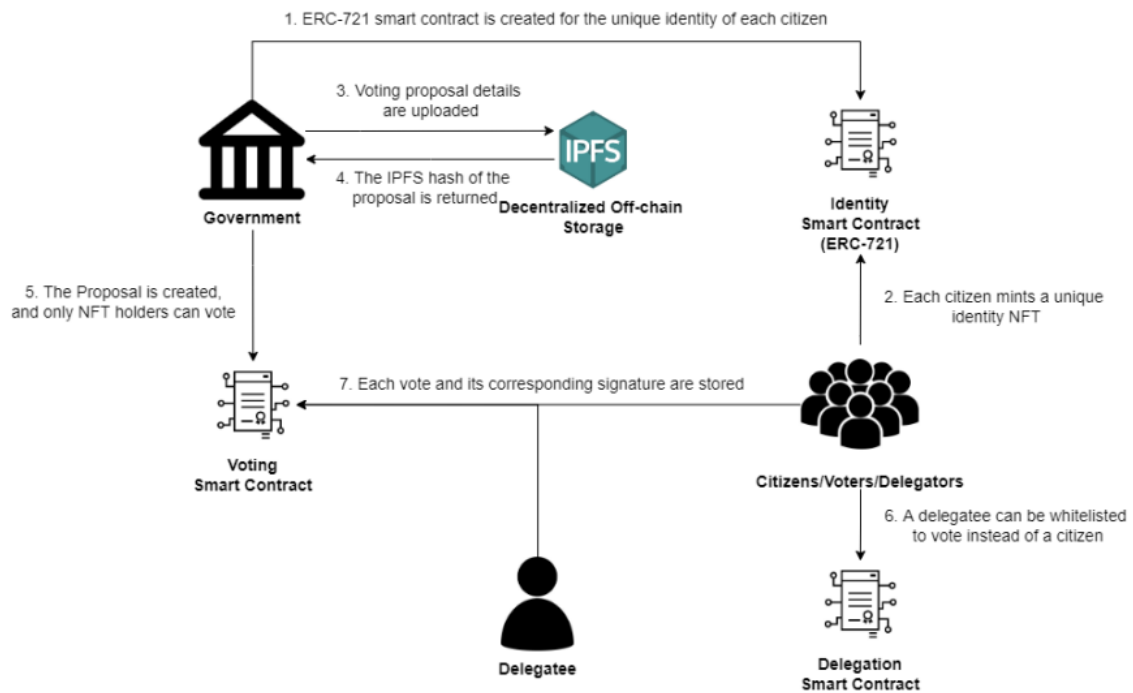


Figure 10: Voting Solution (Musamih, Dirir, Yaqoob, Salah, Jayaraman, Puthal 2022)

6.3 Supply chain

Web 3.0 technologies can have practical uses in various domains, such as supply chain management, where non-fungible tokens can enhance transparency and efficiency, potentially resulting in savings worth billions of dollars.

Efficient and resilient supply chains are crucial for businesses across different sectors, from pharmaceutical giants to local direct-to-customer brands. However, despite

being an integral part of organisations, supply chain networks still need to be more efficient globally (Krishnakumar 2023). The private entity *Tencent Group* is collaborating with the European Union on a new perspective application that involves implementing blockchain and NFTs to enforce Intellectual Property.

6.3.1 Present problems

The WIPO/ACE/15/10 document published by the World Intellectual Property Organization outlines strategies for combatting counterfeit trade. According to the European Union Intellectual Property Office, pirated and counterfeit goods made up 2.5% of world trade and 5.8% of EU imports from third countries in 2019. In 2020, around 33% of EU consumers were uncertain about the authenticity of a product before purchasing, and 40 million EU citizens were tricked into buying counterfeit items. The existence of disjointed tools for controlling supply chains creates gaps that criminal networks can exploit (OECD, European Union Intellectual Property Office 2019).

6.3.2 Obstacles

If we look at the farmer selling his goods to some intermediary for many years, there is a certain amount of trust between them. We can expect high resistance from both sides, even if the new supply chain offers superior value. In addition, corrupt drug supply chain stakeholders exploiting countries such as India and Nigeria would undoubtedly oppose change (Krishnakumar 2023).

6.3.3 Improvements

Transparency

The objective is to employ blockchain technology, enabling any interested party to easily verify a product's authenticity. Therefore, by implementing NFTs, transparency can be achieved, and supply chain management can be improved from the production of raw materials to the display of goods on a website or physical store (OECD, European Union Intellectual Property Office 2019).

Strategy

In addition to transparency, implementing NFT-based procurement and warehousing can provide data scientists with valuable insights into individual product journeys. Detailed data can aid analysts, business owners, and investors in identifying inefficiencies within the supply chain (Krishnakumar 2023).

6.3.4 Achievable solution

To initiate the process, legitimate entities would be granted access to the anti-counterfeiting blockchain via a dedicated portal, enabling them to create digital tokens representing physical goods (digital twins). The rights holders could then authorise third-party entities, such as manufacturers, to handle and create tokens on their behalf and document events and information related to their products. The non-fungible token is recorded on a blockchain and is exchanged between digital wallets as the goods move from one party to another. By combining a distinctive product identity and the ongoing transfer of the digital identity between wallets, it can be demonstrated that the goods are authentic (OECD, European Union Intellectual Property Office 2019).

Customs and other enforcement authorities can access information such as authentic shipping records to support risk assessment as the product moves along the supply chain. Shipping information is stored from the transporter's viewpoint. Although the solution is product-centric, the container can also be tokenised, and algorithms can connect it to the goods it contains. This prevents the need to open a sealed container to check the authenticity of the goods every time it changes hands between parties in the supply chain. Furthermore, tokenised goods with proven authenticity could pass through customs checks quickly (OECD, European Union Intellectual Property Office 2019).

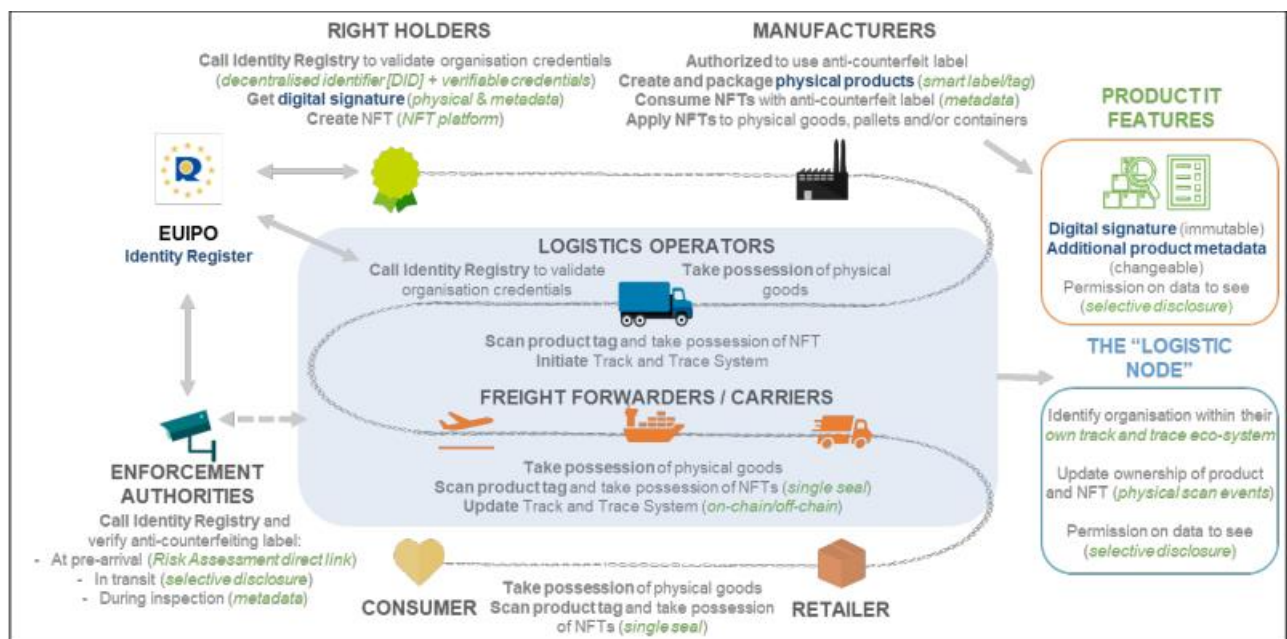


Figure 11: Supply Chain Solution (OECD, European Union Intellectual Property Office 2019)

Conclusions

This thesis aims to explore the concept of non-fungible tokens and their current and potential applications. To achieve this, the study covers foundational topics to provide a comprehensive understanding of the phenomenon, followed by an analysis of the existing market situation and possible use cases to improve the physical world. The research reveals various present use cases being applied, indicating the increasing adoption of NFT technology. The adoption was ultimately confirmed by modelling the current market model, where we can see the ongoing phase the technology goes through. By comparing it with the Gartner Hype Cycle model, the current phase reveals that the hype around NFTs is declining. However, to be able to tell where exactly the technology is and where it is going to evolve, the task was to forecast the future model. The task was fulfilled by using programming software, roughly revealing when the following development phase will happen, and the key finding is that the trend will probably keep decreasing for some time. Moreover, the thesis merges two existing adoption models to highlight the critical step of NFT technology crossing from the early market to the mainstream, marking a positive step for future development. This whole process was essential to indicate opportunities and obstacles for the technology. Finally, the last thing is introducing possible use cases to improve the government sector while divulging present problems. As the technology is still undergoing maturity cycles, obstacles emerge that need to be overcome, and the work underlines them in each use case. In addition, each use case presents an achievable solution for consideration, extending the number of use cases to make the future more manageable for both individuals and organisations.

In summary, this thesis offers valuable perspectives on non-fungible tokens, laying the groundwork for further investigation and progress in this domain. Researchers can leverage the discoveries outlined in this work to deepen our comprehension of non-fungible token use cases and ultimately enhance our collective understanding of this area.

We will monitor this technology's development for years to come, as it will make our relationship with the rapidly changing digital space smoother, eliminating many complexities from the physical world.

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